User Manual: Real Estate Project Feasibility and Financial Appraisal Tool

CE725 Construction Economics and Finance, IIT Bombay. Course Project, Team 8, Topic 2. Version 1.1, 5 November 2026. Team: Maneesha (Project Lead and Finance Modeller), Dinesh Sai (Data Lead), Sujay (AI and Tool Developer), Surya Theja (Validation Engineer), Akshitha (Documentation and Reflection Lead). A longer version with worked explanations of every screen is in `docs/USER_MANUAL_EXTENDED.md`.

1. Purpose and intended users

The tool answers one question: is this real estate project worth committing land and approval money to, and how fragile is that answer? You enter land, construction, approval, sales, financing and tax assumptions. The tool builds a month-by-month cash flow, computes equity NPV, equity and project IRR, payback, profit on cost and the break-even selling price, and shows how each result moves when one assumption changes.

- **Developers and promoters** deciding whether to buy land: use the verdict, peak equity requirement and break-even price.
- **Lenders and investors** judging downside resilience: use project IRR, the debt schedule, the tornado chart and the scenarios.
- **Directors and non-technical readers**: use the plain-language investment memo.

Every number on screen is reproducible from the monthly table and from the companion Excel workbook. The tool is a decision-support aid, not a substitute for professional appraisal; section 7 lists what it does not model.

2. Installation and launch

Needs Python 3.10 or newer; no internet except for the optional Claude memo.

```
python3 -m venv .venv && source .venv/bin/activate    # Windows: .venv\Scripts\activate
pip install -r requirements.txt
streamlit run app.py                                   # opens http://localhost:8501
```

Press Ctrl+C in the terminal to stop. To check the installation run `pytest -q`; 383 tests should pass.

Optional Claude memo. Put an API key in `.streamlit/secrets.toml` as `AIML_API_KEY = "..."` (AIML API gateway) or `ANTHROPIC_API_KEY = "..."` (Anthropic direct), or paste it into the field on the memo tab. The secrets file is excluded from version control; a pasted key is held in memory only.

3. The screen and the inputs

Inputs are in the sidebar on the left in five groups, with a **Reset to benchmark base case** button at the top. The verdict line and two rows of headline metrics sit at the top of the main area, and six tabs below hold the detail: Summary, Cash flow, Sensitivity, Scenarios, Investment memo, Assumptions and validation. Every input change recalculates the whole page in about two seconds. Money is in ₹ crore, areas in sq ft, time in months from project start (month 1 is the first month).

Group	Input	Unit	Base case	Meaning
Timeline and land	Minimum horizon	months	36	Model length; extended automatically if collections run later
	Land cost; payment month	₹ crore; month	20.00; 1	One lump-sum payment
	Approval costs; from, to month	₹ crore; months	3.00; 1 to 3	Spread evenly over the window
Construction	Saleable area	sq ft	1,00,000	Drives construction cost and revenue
	Construction cost	₹/sq ft	3,000	× area = ₹30 crore
	Construction starts, ends	month	4 to 30	Spend follows an S-curve
Sales and collections	Selling price	₹/sq ft	7,500	× area = ₹75 crore gross revenue

Group	Input	Unit	Base case	Meaning
	Launch, sold out by	month	7 to 30	Area booked evenly across the window
	Collected at booking; balance over	%; months	20; 6	Rest arrives in equal monthly instalments
	Marketing and brokerage	% of collections	2.0	Paid when cash is collected
Financing	Loan share of spend	%	50	Of each month's land, approval and construction spend
	Interest rate	% p.a. nominal	12.0	Opening balance × rate ÷ 12, paid in cash monthly
	Collections swept to repayment	%	50	Residual repaid as a bullet in the final month
Tax and appraisal	Income tax on project profit	%	25	Paid in the final month
	Discount rate	% p.a. effective	15.0	Monthly rate = (1 + r)^(1/12) − 1

Input rules. End months must not precede start months, percentages must lie in 0 to 100, months must be 1 or more. A broken rule shows a red message naming the input and stops the calculation. **Horizon.** You set a minimum only; if the last sale is in month 30 with 6 instalments, the model uses 36 months so no revenue is lost. The horizon used is shown in the second metric row.

4. Reading the results

The verdict is **Viable** when equity NPV is positive at the discount rate. The other metrics say how comfortable the margin is. Base-case values let you check your installation.

Metric	Base case	How it is computed	How to read it
Equity NPV	₹7.45 cr	Σ equity CF ÷ (1 + m)^t, month 1 discounted one full month (Excel NPV convention)	Positive: earns more than the hurdle after funding costs
Equity IRR	47.2% p.a.	Monthly rate giving zero equity NPV, annualised (1 + i)^12 − 1	Compare with the discount rate
Project IRR	34.2% p.a.	Same on the unlevered project cash flow	The lender's view of the project itself
Payback	27 months	First month cumulative equity CF reaches zero	How long promoters' money is at risk
Profit on cost	32.7% pre-tax	(Revenue − total cost incl. interest) ÷ total cost	The developer's margin; post-tax on the Summary tab
Break-even price	₹6,250/sq ft	Price at which equity NPV = 0, by bisection	Price cushion before the hurdle is missed
Cost-covering price	₹5,686/sq ft	Price at which pre-tax profit = 0	Below this the project loses money
Peak equity requirement	₹12.86 cr, month 8	Most negative cumulative equity CF	Cash the promoters must have, and when
Peak debt; total interest	₹12.08 cr; ₹2.02 cr	Highest closing loan balance; Σ interest	Facility size; cost of leverage

Cautions. IRR shows n/a when cash flows never change sign or no rate between −50% and +100% per month gives zero NPV. If the equity cash flow changes sign more than once (typically a large bullet repayment) more than one IRR exists; the tool marks the IRR with a warning and you should decide on NPV alone. A high equity IRR with a thin NPV usually means a small equity base, not a safe project.

Cash flow tab. One row per month, inflows positive and outflows negative: land, approvals, construction, marketing, sales booked, collections, tax, project CF, the loan schedule (opening, drawdown, interest, repayment, closing), equity CF and cumulative equity CF. Project CF = collections − spend − tax; equity CF = project CF + drawdown − interest − repayment. **Download cash flow (CSV)** exports the table for an independent check.

5. Sensitivity, tornado and scenarios

One-at-a-time table. Each row changes a single input and holds the rest: price and the three costs at −20% to +20%, sales launch and construction delay at −6 to +6 months, interest and discount rate at −2 to +2 percentage points. Radio buttons switch the metric between equity NPV, equity IRR, project IRR and profit on cost.

Tornado chart. Bars show the metric at ±10% for prices and costs, ±3 months for delays, ±2 points for rates, widest bar at the top. Orange is the side where the metric falls, blue where it rises. In the base case the order is selling price, construction cost, sales launch delay, land cost.

Scenarios. Downside: price −10%, construction cost +10%, launch delayed 6 months (NPV −₹1.88 cr). Severe downside: price −20%, cost +15%, launch delayed 12 months, interest +2 points (NPV −₹11.01 cr). Upside: price +10%, cost −5%, launch 3 months earlier (NPV ₹14.98 cr). Five sliders build a custom case; the chart compares its cumulative equity cash flow with the base case.

6. Investment memo

The memo is a one-page note in seven parts: recommendation, the project, funding, cushion and break-even, key risk drivers, stress test, limitations. **Template mode** (default, offline) fills fixed sentences with the computed numbers and is fully deterministic; use it for any figure that will be quoted. **Claude mode** sends only the fact sheet shown in the expander (inputs, results, tornado and scenario figures) to a Claude model through the AIML API gateway or Anthropic directly, and asks for plain language. The tool then checks every number in the reply against the fact sheet and lists any that do not match in a yellow warning; green means all matched. A matched number can still sit in the wrong sentence, so the memo is a draft: read it against the metric rows before sending it. The person who signs the memo owns the numbers. Do not use Claude mode for a confidential project unless the organisation permits sending the fact sheet to a third party.

7. Key assumptions, data sources and limitations

Conventions (the engine and the workbook both follow them; any hand check must too):

1. Every cash flow occurs at the end of its month; PV = CF ÷ (1 + m)^t with m = (1 + annual)^(1/12) − 1 (1.1715% for 15%).
2. Land is one payment in the land month; approvals are spread equally across the approval window.
3. Construction follows a smoothstep S-curve: cumulative share after k of D months is $3x^2 - 2x^3$ with x = k ÷ D.
4. Sales are booked in equal value each month; each booking collects the booking share immediately and the balance in equal instalments; marketing is a share of each month's collections.
5. Drawdown = loan share × that month's land, approval and construction spend. Interest = opening balance × rate ÷ 12, paid in cash, not capitalised. Repayment = min(opening + drawdown, sweep × collections); any residual is a bullet in the final month.
6. Tax = rate × (revenue − land − approvals − construction − marketing − interest), paid in the final month, zero if profit is negative.
7. Profit on cost = pre-tax profit ÷ total development cost including interest. Break-even prices are found by bisection between ₹1 and four times the entered price with all else fixed.
8. Horizon = the longest of the minimum entered, last sale month plus instalment months, construction end, approval end and land month.

Where the base case comes from. It represents a mid-segment project of 1,00,000 sq ft in the peripheral Mumbai Metropolitan Region (Kalyan-Dombivli or Panvel). Each input was checked against published sources on 5 November 2026; the full table with links is `docs/DATA_SOURCES.md`.

Input	Base case	Published reference
Selling price	₹7,500/sq ft	Knight Frank H2 2025: Mumbai average ₹8,856; Panvel average ₹7,835; Kalyan registered rate ₹7,900
Construction cost	₹3,000/sq ft	Mumbai standard quality ₹3,086 (Oct 2026); JLL 2026 commercial high-rise ₹3,900 to 4,900
Land cost	₹20 cr	JLL: MMR transacted land about ₹17 cr per acre in 2024
Approval costs	₹3 cr	MCHI: 23% to 32% of project cost inside Greater Mumbai; far lower outside BMC limits
Interest rate	12%	RBI repo 5.50% (Oct 2026); NBFC construction finance 11% to 14%

Input	Base case	Published reference
Tax; discount rate	25%; 15%	Section 115BAA 25.17%; EY/NSE 2024 real estate cost of equity 16.4%
Booking 20%, 6 instalments		RERA s.13 caps the pre-agreement advance at 10%; construction-linked plans run 18 to 30 months

Three inputs lean optimistic for Mumbai: the 6-month instalment pattern, 2% marketing, and the approval cost if the site is inside Greater Mumbai. They were kept so the validated benchmark stays intact; a real appraisal should replace them with site-specific figures.

Not modelled. Inflation and price growth; GST, stamp duty and RERA escrow rules; unsold inventory, cancellations and defaults; interest capitalisation, fees, moratoria and lender covenants; phased launches or mixed unit prices; depreciation, loss carry-forward and MAT; probability distributions (sensitivities are one-at-a-time, scenarios are hand-picked). A construction delay only stretches the S-curve, with no overhead or escalation cost, so a pure construction delay can raise NPV slightly (+₹0.36 cr for 3 months in the base case); read that as a model limitation, not a benefit of delay.

Validation. `benchmark/base_case_benchmark.xlsx` rebuilds the base case in plain formulas with no value copied from Python; `python benchmark/compare.py` recalculates it and prints a discrepancy table. Goal Seek on Summary!B2 (NPV) and B14 (profit), changing Inputs!B13 (price), reproduces the two break-even prices. The hand calculation of months 1, 8 and 36 and the NPV sum is in section 6 of the validation report.

Troubleshooting.

Symptom	Cause	Fix
Red message, no results	An input rule is broken	Correct the named input
IRR n/a or marked with a warning	No sign change, or more than one sign change in the equity cash flow	Decide on NPV; raise the sweep to shrink the bullet repayment
Break-even price n/a	No price up to four times the entered one gives zero NPV	Lower costs; the project is far from viable
Claude memo fails, or yellow number warning	No key, no internet, rate limit; or the model used a figure not in the fact sheet	Use template mode; check each listed number against the metric rows
Workbook differs from the app	Not recalculated, or an input differs	Press Reset in the app; force recalculation (Ctrl+Shift+F9)

For questions about the model contact the Project Lead and Finance Modeller; for installation problems the AI and Tool Developer.